CHARUSAT®

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY AND ENGINEERING

DEVANG PATEL INSTITUTE OF ADVANCE TECHNOLOGY AND RESEARCH

DEPARTMENT OF COMPUTER ENGINEERING

A.Y 2026 – 27 Semester – 3

LAB MANUAL

CSUC201 Fundamentals of Data Structures and Algorithms

DEPSTAR

Practical 8: Binary Tree Traversals and Binary Search Tree Insertion

Practical Aim :

To implement binary tree traversals and binary search tree insertion, and develop the ability to reason about how the order of visiting nodes changes the information retrieved from a tree..

Problem 1 :

A company stores its organizational hierarchy as a tree — the CEO at the top, managers below,

and employees at the leaves. Different departments need the hierarchy printed in different

orders: HR wants each manager printed between their left and right sub-teams, the archive

department wants each manager printed before their sub-teams, the payroll department wants

each manager printed after all their sub-teams are listed, and the floor manager wants everyone

printed level by level from top to bottom. Given a binary tree representing the hierarchy,

implement all four printing orders and display the result of each.

Describe how your solution for level-by-level printing differs from the other three. What

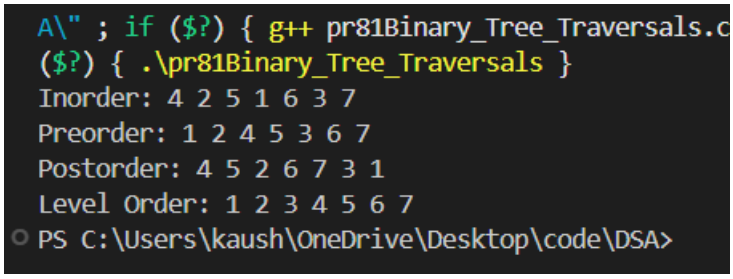
additional structure did you need to use, and why does the same recursive approach that works

for the other three not directly apply here?

For Code:

https://github.com/KaushalPatel22/FDSA/tree/main/FDSAGITSUBMITION

Output:



Problem Faced:

Understanding different tree traversals and implementing level-order traversal using a queue.

Outcome/Conclusion:

Successfully implemented all four tree traversals: inorder, preorder, postorder, and level-order.

Problem 2 :

A school library uses a system where every book is assigned a unique code when it arrives.

Books are organized so that at any shelf position, all books to the left have smaller codes and

all books to the right have larger codes. This arrangement lets staff find any book quickly by

always knowing which direction to look. Given a sequence of book codes arriving one by one,

insert each into the correct position and print the inorder sequence after all insertions are

complete.

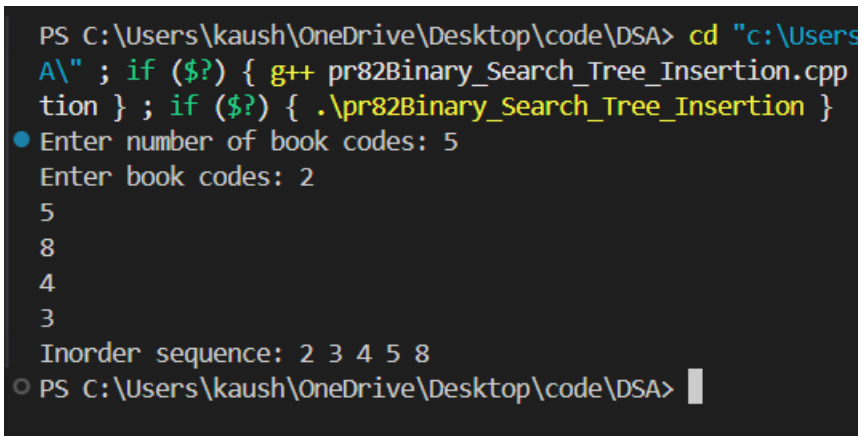
Describe how your solution decides where to place each new book code. What property of the

final inorder sequence tells you whether all insertions were performed correctly?

For Code:

https://github.com/KaushalPatel22/FDSA/tree/7b757e98950f9003ab582799d5dc37f284b72f8e/FDSAGITSUBMITION

Output:



Problem Faced:

Inserting nodes at the correct position in a BST while maintaining its properties.

Outcome/Conclusion:

Successfully implemented BST insertion and verified the sorted output using inorder traversal.

Key Questions / Analysis / Interpretation to be Evaluated During/After Implementation

1. [Problem 1] Preorder traversal always visits the root before anything else. Using only this property — without tracing any tree — explain whether it would be possible to reconstruct the original tree from its preorder output alone, and what additional information you would need if it is not.

ANS :

Preorder alone cannot uniquely reconstruct a tree. Inorder traversal or additional structural information is needed.

2. [Problem 2] The inorder traversal of a correctly built BST always produces a sorted sequence. If your inorder output is not sorted after all insertions, what does that tell you about where the error is — in the insertion logic, the traversal logic, or possibly both? How would you isolate which one?

ANS :

An unsorted inorder output indicates an error in insertion or traversal logic. Check the BST structure and verify the inorder traversal separately.